



# Examiners' Report Summer 2026

Pearson Edexcel AEA Mathematics

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## General Comments

The paper provided a fair representation of topics from across the A-level specification including the statistics and mechanics sections, but it was on the latter of these that learners most struggled. Questions 1, 2, 3 and 7 all provided a good source of marks, the calculus in question 7 being one of the traditional strong points. But questions 4, 6 and in particular question 5 each posed a significant challenge to discriminate between merit and distinction.

There was, as usual, some excellent application of the expected methods as well as some interesting alternative approaches, highlighting the skills and problem-solving ability of many learners. Weaker responses saw learners unable to progress into the deeper parts of questions, particularly in the middle questions, with some erratic work to try and find solutions at times.

Calculus remains one of the stronger topics for learners taking this paper, and featured in many of the questions to some degree, although question 6 did discriminate well on this topic. The statistics was well approached, but the mechanics question was disappointingly poorly handled.

## Individual Question Report

### Question 1

The first half of this question was accessible to most learners as it was testing basic application of differentiation from first principles and the trigonometric addition formula. Learners were familiar with the process of finding an expression for the gradient of the chord and taking limits. Most were able to put the terms over a common denominator and realised they needed to use the addition formula in the numerator and did so correctly, with most scoring the first three marks. However, many also applied the  $\sin(x + h)$  formula to the denominator and sometimes multiplied through by  $\sin x$ , making the denominator unnecessarily cumbersome.

The next three marks were more discriminating, with many not able to clearly exemplify the method, although most shown an understanding of the correct general procedure. Those who made clear the separation of the terms to identify the limits usually proceeded to score full marks, but many applied the two given limits in situ without clear demonstration of them and so could gain the B mark only of the last three. Some were not able to factorise to the form where standard limits could be used at all, and others went through multiple rearrangements instead of focusing on how to actually extract the given limits.

There were a few cases of learners trying to use small angle approximation instead of using the standard limits provided in the question, and a few tried using (in some cases trying to prove by first principles first) the quotient rule and derivative of  $\sin x$ . These were not permitted credit for the method but could access the B mark if limits were applied appropriately. Overall, the trigonometric manipulation and factorisation proved to be the hardest part of this question.

Most learners used the limit notation very well throughout their proof even if the extraction of the limits was not clear.

## Question 2

This question again provided good early access with most learners able to score marks in parts (a) and (b) but part (c) was a good discriminator with only the stronger learners producing full marks, and many unable to access either mark in part (c).

In part (a), most learners completed the square and deduced the correct value of  $k$ . Differentiation was also a common alternative method and also carried out well. However, some learners left their answer as  $x = \frac{3}{2}$  rather than the correct answer which is  $k = \frac{3}{2}$ , and so learners should be careful to check what they are being asked for and answer appropriately. Some leniency was given for those who stated the full domain, but alone  $x = \frac{3}{2}$  could not score the accuracy.

Part (b) was accessible with most learners gaining the first method mark by attempting to make  $x$  the subject of their completed-square form, or by first swapping  $x$  and  $y$  and then making  $y$  the subject. A few used the quadratic formula rather than completing the square, also generally successfully in reaching the correct form. However, many did not obtain the correct inverse function, commonly selecting the incorrect root or failing to select a root at all and leaving the  $\pm$  in their answer. A few learners used the incorrect notation  $f'(x)$  instead of  $f^{-1}(x)$ , but overall the function notation was used well. The most common error in finding the domain of the inverse function was giving the answer as  $x > \frac{11}{4}$  rather than  $x \geq \frac{11}{4}$ .

Learners found part (c) more challenging and many learners scored no marks here. Attaining a correct domain was more common than a correct function, but both aspects caused difficulty. The most popular incorrect approach was to replace  $x$  with  $x^2$ , rather than taking the square root of the expression obtained in part (b). Others tried working through from a restart with  $h(x)$ , which had better success in reaching a correct form, but the choice of signs was often incorrect, and again many left their answer with  $\pm$  in the equation, sometimes in both places. Domains were more erratic, though the lower bound was more successfully achieved (sometimes via follow through) than the upper bound, and various incorrect upper bounds were seen.

### Question 3

This question was pleasingly well attempted, perhaps helped by a recent question on a similar theme on a previous paper. Learners showed a good understanding of probability distributions and realised the need to sum the geometric series to prove the results in (b) and (c).

Part (a) was very well answered showing that learners understood the definitions of the distribution in both the even and the odd cases, with only a few weaker responses failing to make any headway. Learners demonstrated high confidence in establishing the next two probabilities in the distribution.

There was also good access in part (b), which was efficiently done by the many who understood that the number of possible cases was infinite. Most learners understood the requirement for the sum of all the probabilities to be 1 and recognized the infinite nature of the progression of both odd and even terms. They applied the sum to infinity formula confidently, and either summed the two expressions, or in more efficient cases subtract the sum of the odd term probabilities from 1 first before simplifying. The algebraic manipulation that was then needed was generally carried out successfully, although some learners made harder work of it by trying to rationalise the denominator of the sum of even terms before summing, and then having to repeat the procedure after rearranging.

Some learners, however, assumed that the number of cases was finite (usually 8) and were not (sometimes after a great deal of arithmetic and algebra) able to access the two marks for this part.

Part (c) was less successfully done and acted as a good differentiator, with only the more mathematically precise learners securing full marks. Those who assumed a finite number of cases could score at most one mark, but they often displayed a lack of understanding of what was required even in finding the probability of selecting an even number less than or equal to eight.

A few learners just worked out the probability of an even number less than or equal to 8. They scored only one mark, if successful. More successful learners knew that they had to divide this probability by the probability of getting any even number. Many learners made the resulting algebra and arithmetic more difficult for themselves by overlooking that the probability of any even number was  $1 - \frac{3}{8} = \frac{5}{8}$  where the  $\frac{3}{8}$  was found in the previous part. Such learners often worked with their irrational expression in terms of  $a$  from the sum of even terms  $\left(\frac{a\sqrt{5}}{\sqrt{5}-1}\right)$  and struggled to simplify the resulting fractional expression, being unable to successfully reach a rational value, meaning they still only secured 1 mark overall for this section.

Learners were more secure in part (d), with many able to enumerate the cases for  $Y = 3$  and for  $Y = 4$  followed by the correct probabilities and then their sum. A few

learners left their answer as the sum of two fractions or as a single fraction not in its simplest form and they were not awarded the final mark. The most common errors here were to omit a case (often considering only the case where  $Y = 4$ ) or to include  $Y = 5$  as an extra case.

## Question 4

This question is where the difficulty of the paper ramped up and learners had to answer an extended question. Though part (a) was generally well attempted and gave access to many learners, part (b) was much more discriminating and only relatively few stronger responses scored heavily in this question.

Part (a) (i) was an accessible start with most realising the curve and circle being tangential meant need to show a single solution for either  $y$  or  $x^2$ . A less common approach was to use that the gradients of the normals needed to match, but this also was effectively applied by some. Although not the most efficient, the commonest approach was to substitute  $y = Ax^2 + k$  from the equation of the parabola into the equation of the circle and then apply the condition for repeated roots to the resulting quadratic in  $x^2$ . After some manipulation the given expression for  $k$  was usually found. A few learners used the more efficient way of substituting for  $x^2$  to get a quadratic in  $y$  in which the resulting algebra was more straightforward. Those matching the gradients of the circle and the parabola would get the point of contact  $y = -\frac{1}{2A}$ , and could then find  $x^2$  at the point of contact using the equation of the circle and hence the value of  $k$  from the equation of the parabola.

Some learners who could not see how to proceed from the given set up instead tried to work backwards using the expression in  $r^2$  from part (iii) to work out the value of  $y$  in (ii) or  $k$  in (i). Such responses were not credited as they had not established the results for themselves.

Another common errancy was to state that the discriminant was positive (or  $\geq 0$ ) before switching to an equality at the end of the manipulation in unconvincing manner, precluding the award of the final mark.

Parts (ii) and (iii) were usually well done, although some responses took a very long and circuitous route to arrive at the final coordinate  $y = -\frac{1}{2A}$  for (ii). While these long methods ultimately yielded the correct results, they increased the potential for algebraic arithmetic errors. More elegant, direct simplifications would have been preferred and would have saved considerable time.

Responses to part (a) (iv) were less successfully completed and proved to be a subtle discriminator, as the strict inequality on  $r$  often missed. Errors included giving ' $r \geq \frac{1}{2A}$ ' or the combination ' $r > \frac{1}{2A}$  and  $r < -\frac{1}{2A}$ '. Both of these errors contradict the original information given.

Part (b) proved to be a very good discriminator across learners, as few recognized the underlying idea was one of translation of the two curves  $C_1$  and  $C_2$  by means of the vector  $\begin{pmatrix} a \\ b \end{pmatrix}$ . (Hence,  $x^2 + y^2 = r^2$  becomes



$(x - a)^2 + (y - b)^2 = r^2$ ). Those who understood this usually completed (i) in two lines, replacing  $x$  with  $x - a$  and  $y$  with  $(y - b)$  before substituting for  $k$ , but there were many who simply did not attempt this part at all. Of those that did understand the requirement the A mark was often lost due to sign or expansion errors made before substituting for  $k$ . Another more cumbersome but potentially successful method was to compare coefficients of the quadratic, but the same accuracy errors in expansion would occur in such responses too.

For part (ii) alert learners realized that translation preserves parallelism so they could work with the normals using the  $x$  and  $y$  values found in part (a). Since the value of  $y$  in part (a) was already given and it was a simple manipulation to find the expression for  $x$  these learners derived the correct equations very quickly, but some missed out the normal parallel to the  $y$ -axis – i.e.  $x = 0$  translated to  $x = a$ .

However, such responses were in the minority here, with many not even offering an attempt, and of those the route taken was generally not the most optimal approach, with learners tending to work with the new centre and coordinates to find the gradients and equations rather than realise they could work with the original circle and parabola and translate later. Use of derivatives rather than recognising the line was a radius of the circle was also common, which frequently led to algebraic errors.

This was the first question on the paper which carried an S mark – given for efficient methods in (a)(i) and (b)(ii), for example, or other succinct approaches or good style and communication shown. Learners should always be on the lookout for alternative, possibly simpler, methods when they solve problems.

## Question 5

This question proved challenging for many learners, with very few achieving a substantial tally of marks. The standard of work throughout this question was disappointing with learners often presenting muddled and uncertain work, and the question performed below its expected profile. The presence of such a question should not be a surprise to learners and yet many seemed unprepared for it and particularly had difficulty with the justifications in part (c). The challenge and strength of this question, in particular part (a), was the openness with no structure which gave freedom for learners to exhibit their aptitude but also meant that those who were not able to get started had little hope of continuing to the end.

The overriding issue in part (a) that hampered many learners was a lack of clearly resolved systems of forces for the two blocks. Although, most learners were able to resolve all forces acting on blocks A and B this was often only tacit in their working, with the perpendicular equations only being apparent by virtue of the substitution into the maximal friction equation. Many also tried to “resolve along the slope” for both blocks combined, omitting the tension entirely, and making it difficult to determine if they had truly resolved both systems in their entirety. Those who clearly resolved both systems first are the ones who fared better on the question as a whole, as they were able to identify the directions for the friction and components for force  $P$  more accurately to get the two extremes set up correctly.

Use of the diagram in the question was of help, with many students annotating to show the force diagrams, though very few actually sketched out a forced diagram in their working, something that would have perhaps given them a better understanding of the problem.

A common error was to assign friction in the incorrect direction, though these could often score the first five or six marks before missing out on the second extreme case. Some responses omitted the force components of  $P$  from the equations for block B entirely (forfeiting most marks). The most secure mark was the third method mark for using limiting friction in the equations parallel to the plane, although the use of the inequality  $F \leq \mu R$  was rarely seen and equality used.

Very few learners recognised that  $T$  must be zero for the upper bound, meaning the final two marks were lost by most. Learners often did not clearly show their strategies to find the extreme values of  $k$ , relying on the answers they found to imply the marks on some occasions. Such responses were automatically forfeiting the right to any 5 marks in the question as the solutions were far from clearly communicated.

Part (b) also was found surprisingly difficult by learners with many bypassing the part, though no evidence that it was overlooked at the bottom of the first page of the question as parts (c) and (d) were correctly labelled. Rather, learners

simply appeared not to know how to find the reaction force on a pulley, indeed, of those who did answer many attempts resolve only horizontally and neglected the vertical components, a few resolved just the vertical components. Relatively few learners used the cosine rule and addition formula to determine the resultant force on the pulley. Where this method was used, it tended to be more successful than resolving the force into horizontal and vertical components and then applying Pythagoras' theorem. Even so, there were a surprising number of algebraic slips in the process, such as forgetting to square root at the end and given  $R = \frac{64}{65}T$ .

Parts (c) and (d) required learners to explain the effects of decreasing and increasing  $P$ , respectively, but very few convincing explanations were given. Although there was some success in identifying the effects in part (c), learners often did not provide reasoning that was sufficiently clear or rigorous to justify all of them. Often just one or two marks were scored for one correct effect and reason, with the special case applying in part (c) where effects were correct but no sufficient reasons. In part (d) very often, due to the lack of understanding shown in part (a), the conclusion would be that block A moves down the slope, and tension decreases, rather than being zero, so the special case seldom applied here.

Overall, while learners had some idea about the general process to get started in part (a), very few were careful enough to resolve both systems fully for both cases and, even though these explanations focussed on the effects and could have given an opportunity to correct some earlier oversights, the necessary situational awareness was not exemplified by many to do so. Very few S marks were awarded as a result, and those who did well enough to achieve S marks in this question were by and large distinction level learners.

## Question 6

Although calculus is generally a strong topic for learners taking this paper, this question did test understanding and discriminated well. While not as low access as question 5, it did prove to be the second most challenging question on the paper in terms of performance. Overall, this was a challenging question for most, but not necessarily due to lack of skill but due to not fully understanding the flow of the question.

Part (a) was easily the most successfully answered part in this question, being a fairly staple A-level type question to ease into the rest of the question. Most learners made scored both marks, most commonly by making  $x$  the subject, differentiating implicitly, taking the reciprocal and using the identity  $\sec^2 y = 1 + \tan^2 y$  to rewrite in terms of  $x$ . A few converted to  $\cos^2 x$  first and used a triangle with Pythagoras to derive the result, and only few score no marks here.

There was less success in part (b), more due to a lack of care than a lack of understanding. Many learners failed to appreciate that the demand of the question was to show the result using the given substitution and were too informal in reasoning. A lot of responses missed one or both marks because of this, even though the learners clearly possessed the correct skills. Most made an attempt at the substitution and while many did clearly demonstrate all four bullet points and complete with convincing argument to achieve the given expression, there at least as many who did not complete successfully. Achieving  $dx = -du$  or equivalent was the most successful of the bulleted point achieved, the other three had mixed success rates. Some simply replaced  $x$  with  $u$  in the integral rather than with  $a + b - x$ , with or without changing limits, while others neglected to change limits. Knowing to change the sign when switching the limits was shown well by most, but those who did not change limits initially could not show this well. Learners who started with one of these false steps would usually somehow complete to the given result, but had already lost the marks for the unconvincing argument.

Part (b)(ii) was much better answered, as long as it was attempted. The result was easily deducible even for learners who were not able to do other parts, although some learners had several rows of working to achieve this.

Part (c) was found more challenging by learners, and the discriminating aspects of the question were more apparent here, with again many convincing and many less convincing responses seen. Surprisingly many learners missed the connection between parts (b) and (c) and simply tried to argue that

$\int_{-\pi}^{\pi} \frac{x}{1 + \cos^2 x} dx = 0$ , which did not score any marks unless they were awarded the special case for arguing that from the oddness of the integrand. For those that did use part (b) success was reasonable, however, most of these used the result from part (b)(i) instead of part (b)(ii) which meant that more work had to be done;

adding two of the original integral (2I), substituting  $-x$  for  $x$  in one of the terms (applying (b)(i)), simplifying to achieve 2x the given expression. Where attempted this was done well, but those who used part (b)(ii) made much lighter work of it and were generally from stronger responses. A few attempted to (incorrectly) integrate  $\int \frac{x}{1+\cos^2 x} dx$  and show that when the limits were substituted the limits it evaluates to zero, but these weaker responses gained no credit.

There was greater success with part (d) and again many were able to access a mark here when other parts were left out or poorly answered. Learners showed familiarity with odd and even functions and argued this part well enough for the one mark on offer. A few learners argued that  $\cos(-x) = -\cos(x)$ , which cost them the mark even though the result was still correct, but aside from this the result was generally successfully demonstrated.

Finally, part (e) again was a good discriminator for strong learners, with many weaker responses failing to make progress beyond the first mark if even that. Once learners made progress into the second mark success in the remainder increased dramatically with many then being able to go on to achieve full marks. To get started learners needed to apply their knowledge of integrals of odd functions, but many instead attempted to use some kind of substitution, or to try integration by parts, and end up making no suitable progress. There were a few cases where a substitution led to an equivalent integral for the second half of the problem being carried out, but also with another arctan integral to evaluate, but most stopped at this stage and so no credit was achieved. Those rare cases who did argue successfully from symmetry that the remaining integral disappears were given some credit, but the question specified to use earlier parts to achieve the result, so credit was only partial.

Those who did identify the properties of the odd function were generally able to access the first mark, but again at this stage there were many who either stopped or proceeded to make false attempts at the integral. Those that were able to apply the results from parts (b) and (c) generally achieved the second stage integral required, although again there was generally a lot of excess working that went into this, as the most efficient approach was often missed and learners reverted to using part (b)(i) and essentially repeating the whole process of part (c) again.

Those who got to this point were generally able to use a suitable substitution and integrate to  $\arctan u$  and apply the correct limits or apply the reverse chain rule process using part (a). Aside from a few who made slips (evaluating  $\arctan(1)$  incorrectly to  $\frac{\pi}{2}$  being surprisingly common), most who reached this stage completed successfully to achieve the correct answer.

For those who completed fully most were able to gain at least one of the 5 marks, though it was less common for both to be awarded. This was most

commonly awarded for the concise work dealing with  $\sin(\pi - x)$  and  $\cos(\pi - x)$  or for using the changed limits.

## Question 7

Considering that this was the final question on the paper, a large amount of the question proved to be accessible, and a lot of learners were able to make a reasonable attempt at most of the question. The structure of the question very much led them through and armed them with sufficient information to be able to show that  $g(x)$  is an increasing function. As has been noted, calculus tends to be the strongest topic for learners taking this paper and this proved to be the case here. Although very few managed to achieve full marks, it was only due the very demanding final mark, and there were many excellent responses that came close to full marks, with even the S marks being accessed with some regularity on this question.

Part (a) was completed well by most learners, with most using implicit differentiation efficiently and accurately. Those who did not make progress in this part would generally not get any further than this point in the question.

Success was slightly less secure in part (b), but many learners were able successfully complete it. The unsuccessful attempts tended to be due to trying to apply the chain rule directly to the function rather than taking  $\ln$  of both sides first, and so did not reach a suitable expression, but rather something akin to  $\frac{dy}{dx} = 2x \times x^x (\ln x + 1)$ . Those who did recognise the need to repeat the process of taking  $\ln$  generally reached an expression for  $\frac{1}{y} \frac{dy}{dx}$  before multiplying back across. The first two methods of the scheme were both seen regularly, but the third approach was less common. However, each approach was carried out well when seen. Most efficient were those who spotted that they could write  $(x^x)^x$  as  $x^{x^2}$ . The most common error was the loss of the factor 2 before  $x \ln x$ . Few spotted that they could set  $\frac{dy}{dx} = 0$  earlier and proceeded to substitute back in for  $y$  before continuing to solve.

Having scored the first 2 marks, learners were generally able to obtain the correct  $x$  value, but the  $y$ -value caused problems for many, particularly in reaching a simplified form, and some convoluted answers appeared, including with 3 levels of indices.

Part (c) was often completed more successfully than part (b), repeating the process to obtain a derivative (even when this had not been done in part (b)), substituting for  $y$  and manipulating the expression into the required form. The first 2 marks were consequently commonly scored. There were some slips in the subsequent factorisation, often because of over-complicated approaches, but most were able to achieve the correct result, despite that even some of the stronger responses had errors in obtaining the expression for  $Q(x)$ .

There was also generally good progress in part (d) in attaining at least one correct solution, even when learners had struggled in earlier parts, learners spotting that there was a re-entry point here. However, many cancelled out the  $\ln x$  term, thus failing to obtain the value  $x = 1$ , while some others identified this but failed to solve for the  $x = e^{-1}$  solution. It was also common to see  $x = 0$ , but better responses identified that  $x > 0$  was required.

Again, many learners were able to gain marks in part (e) even if earlier parts of the question were less successful. Learners were not fazed by having to differentiate the slightly more unusual  $(\ln x)^2$  term and easily coped with applying the product rule appropriately and solving their resulting quadratic in  $\ln x$  to obtain the values of  $\alpha$  and  $\beta$ . Occasionally these were given the wrong way round and sometimes notation needed more care with floating minus signs making it awkward to discern accuracy. Errors in differentiating  $P(x)$  often led to trivial values, but such responses were less common, the differentiation overall being of a very high standard.

Once again in part (f) the differentiation to find the second derivative was usually correct, and this was the most popular way to determine that  $P(x)$  was minimum  $x = \beta$ , by attempting to find the value of the second derivative at this point. Because most had obtained the correct value of  $\beta$ , they were able to obtain the correct value of  $P''(\beta)$ , show  $P''(\beta) > 0$  and conclude that there was a minimum at  $x = \beta$ . There were some nice responses where learners appreciated that  $x > 0$ , and in particular  $\beta > 0$ , and so obtained an expression  $P''(\beta) = \frac{\sqrt{5}}{\beta} > 0$ . The most common error here was to choose the smaller of the two roots to evaluate at, though this was not common. Slips in the evaluation also occasionally occurred.

While the progress through parts (a) to (f) had generally been very good, part (g) was much more discriminating and there were very few strong attempts at this final part of the question with many learners making no attempt at all. Some indicated that they knew they needed to show that  $g(x) > 0$  for  $x > 0$  and identified that  $x^{Q(x)} > 0$ , but then did not know how to proceed further. Only the stronger distinction level responses showed a good understanding of the requirements of this part.

Some learners made a brief and weak argument that because  $P(x)$  has a minimum at  $x = \beta$ ,  $P(x) > 0$  and so  $g(x)$  is increasing. There were also some learners who considered the curvature, attempting to show that  $P''(x) > 0$  for  $x > 0$  and there were some attempts to argue that  $P(x) > 0$  for all values of  $x$  by considering each term, without considering the value of  $P(\beta)$ , with varying degrees of success. There were, of course, also attempts to find the value of  $P(\beta)$  but fewer rigorous arguments that  $P(\beta) > 0$ .



The final mark was particularly demanding and seldom scored, the full list of relevant points required for the mark not being satisfied. It was the need for  $g(x)$  be continuous and utilising part (d) of the question to argue that  $y = 1$  is crossed only once that were most often missing.

Most of the learners who scored 16 or more demonstrated enough S+ points to gain the S2. Their work tended to be clearly set out and with evidence to support any assertions. For those scoring 13–16 success in achieving a single S mark was less certain, but many did exemplify sufficient S+ marks to access it even if their work was not succinct.

## Summary

Learners should

- be sure to read questions carefully to know what is asked for and to appreciate the holistic flow of a question where earlier parts lead into later parts
- set out work clearly with good logical flow throughout, and especially to maximise the scoring of S marks
- avoid trying to do too many steps in one go and thus obscure key elements of the working anticipated